

LLMs for corpus annotation: Close and extended replications of Morin & Marttinen Larsson (2025) using smaller open LLMs and BERT

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1 Abstract

Large Language Models (LLMs) are increasingly being leveraged for the annotation of large corpora. In an early evaluation of such an application, Morin & Marttinen Larsson (2025) presented “a replicable method for automating grammatical annotation” using Anthropic’s model Claude 3.5 Sonnet (Morin & Marttinen Larsson 2025: 4). Their annotation pipeline consisted of a complex “conversation” with the model over several iterative phases of training and testing. They were able to achieve an accuracy of 93% compared to the gold standard provided by a human annotator (Morin & Marttinen Larsson 2025: 6) and conclude that the pipeline may also be effective for other linguistic tasks in other languages (Morin & Marttinen Larsson 2025: 7).

We aimed to repeat Morin & Marttinen Larsson (2025)’s “replicable method” as closely as possible, using the same data, albeit with two open LLMs, Mistral Small 3.2 24B Instruct (MistralAI 2026) and OpenAI GPT OSS 120B (OpenAI 2026). While the smaller Mistral model did not perform as well on the task (accuracy of 67%), the open GPT model yielded very similar results (accuracy of 94%). Although the authors published their data and a full transcript of their “conversation” with the model (Morin & Marttinen Larsson 2024), we found that essential information to independently replicate the study (e.g., whether the web interface or an API was used, analysis code, etc.) was missing - forcing us to make some educated guesses as to the exact procedure. Our results suggest that, as of early 2026, large commercial LLMs may not be necessary to complete this kind of linguistic task in contemporary English.

Next, we tested the need for the extensive ‘in-conversation’ training process outlined in Morin & Marttinen Larsson (2025). To this end, we conducted a second, extended replication applying

a much simpler few-shot training approach with the best-performing open LLM from our first replication. Our results suggest that the training process need not be as elaborate as in Morin & Marttinen Larsson (2025) to accomplish similarly high accuracy rates.

Finally, we replicated the annotation study using a fine-tuned DistilBERT model (Sanh et al. 2020). Although this encoder-only transformer is considerably smaller than even a small generative LLM, it performed approximately on par with the 24B Mistral model used in our first replication. These findings indicate that task-specific supervised models may provide a competitive and certainly more computationally efficient alternative to generative LLMs.

Our results have important implications for the replicability of corpus-linguistic research involving LLMs as part of the annotation process. We argue that, when choosing a method, issues around model accessibility, data leakage, and carbon footprint can and should be taken into consideration.

2 Introduction

Over the past decades, linguists have gained access to ever larger corpora of natural language. This poses great opportunities, but also considerably increases the workload associated with the manual and/or (semi-)automated preprocessing and annotation of large amounts of data. The rise of large language models (LLMs) is therefore an exciting development for linguistics research, offering new possibilities for complementing and extending existing automated methods for processing larger datasets and corpora. In recent years, a growing number of publications have reported on the usefulness of LLMs for a range of linguistic annotation tasks, such as the annotation of speech acts (Yu et al. 2024; Su & Ye 2025), metaphor detection (Fuoli et al. 2026), word sense or function disambiguation (Fonteyn et al. 2024; Morin & Marttinen Larsson 2025), and political text classification and annotation (Heseltine & Clemm Von Hohenberg 2024).

While LLMs are increasingly prevalent in both linguistic research and our everyday lives, they also raise a number of concerns. For research purposes, specifically, their ‘black-box’ nature raises concerns over the transparency, reproducibility, replicability of the research produced with their assistance. Note, that although the terms have been used interchangeably in the past, we distinguish between reproducibility and replicability following the definitions of the FORRT (Parsons et al. 2022) and Turing Way (Community 2022) communities. We thus define *reproducibility* as the ability of an independent researcher or team to obtain the same results as an original study using the original data and methods (see Figure 1). This is in contrast to *replicability*, where the same methods are applied, but on different data; and robustness, where the same data, but different methods are used. Only if an effect can be reliably observed across different datasets with different methods, can we claim that it is *generalisable*.

In this study, we seek to replicate Morin & Marttinen Larsson (2025)’s methodological study, whose title claims to present a “replicable method” for leveraging LLMs for linguistic annotation.

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Figure 1: Term disambiguation (Community 2022, CC BY 4.0)

The original study was conducted using a commercial, closed-source LLM: Anthropic’s model Claude 3.5 Sonnet (Morin & Marttinen Larsson 2025: 4), which is no longer available for use. We therefore first sought to apply the original methodological pipeline to the same data, but using two open-source LLMs, whose parameters can be controlled to generate reproducible results. In an extended replication, we subsequently re-ran the task with a much simpler pipeline than that proposed in the original paper. Finally, we repeated the classification task using a simpler and widely used NLP tool from the BERT family, distilBERT.

While large, cloud-based, proprietary LLMs continue to top performance benchmarks, open-source LLMs have made significant progress in closing this performance gap (Manchanda et al. 2025: 4). We consider open models to form a crucial part of the open research ecosystem (alongside open materials, open data, open code, open documentation, open access, etc.). Openness promotes transparency, reproducibility, and quality control; which are necessary features to support accessible, reliable, and credible empirical findings and build trustworthy AI (Liesenfeld, Lopez & Dingemanse 2023). We also believe that smaller, locally-run LLMs are worth considering for environmental reasons: Training and operating LLMs with billions of parameters requires substantial computational resources, including large amounts of water (e.g. for cooling data centers) and electricity, generating substantial carbon emissions (Özcan et al. 2025; Kron 2026). Although AI becomes cheaper and faster, its global adoption is rapidly increasing resource consumption (Jegham et al. 2025).

3 Research aims and questions

This study aims to replicate the analysis published in:

Morin, Cameron, and Matti Marttinen Larsson. 2025. “Large Corpora and Large Language Models: A Replicable Method for Automating Grammatical Annotation.” *Linguistics Vanguard* 11 (1): 501–10. <https://doi/10.1515/lingvan-2024-0228>.

The data associated with this project is publicly available in an OSF repository (Morin & Marttinen Larsson 2024). It contains a “conversation transcript”, from which the iterative training process could be traced, and several datasets for training and testing an LLM on the task of differentiating between evaluative and non-evaluative uses of the verb *consider*. The model outputs were evaluated against a human gold standard provided by the authors.

We seek answers to the following research questions (RQs):

1. Can smaller, open-source models perform as well or better on Morin & Marttinen Larsson (2025)’s consider annotation task as Claude Sonnet 3.5 based on the accuracy metrics reported in (Morin & Marttinen Larsson 2025: 6)?
2. If RQ1 is answered in the positive, can this also be achieved with a significantly less extensive training process than that proposed in Morin & Marttinen Larsson (2025)?
3. Can a small, encoder-only transformer model from the BERT family, distilBERT, perform as well or better on the same task as Claude Sonnet 3.5 based on the accuracy metrics reported in (Morin & Marttinen Larsson 2025: 6)?

While our close replication examines RQ1, the extended replication serves to answer RQ2 and RQ3. We test the performance of two open-source LLMs: Mistral Small 3.2 24B Instruct 2506 (MistralAI 2026) (hereafter referred to as “Mistral”), an open-weight model released in 2025 under Apache 2.0 license by the French AI company Mistral AI, and OpenAI GPT OSS 120B (OpenAI 2026), an open-weight large language model that was also released under Apache 2.0 license by OpenAI (hereafter “GPT”).

The outcome of the replications will be compared and assessed based on the metrics Morin & Marttinen Larsson (2025) provided in their evaluation: Accuracy, Precision, Recall, F1 score, and Matthew’s correlation coefficient (Morin & Marttinen Larsson 2025: 6). We formulate the following hypotheses:

Hypothesis 1.1: The model Mistral Small 3.2 24B Instruct 2506 performs as well as or better than Claude Sonnet 3.5 on the *consider* task, as measured by the accuracy metrics reported in Morin & Marttinen Larsson (2025).

Hypothesis 1.2: The model OpenAI GPT OSS 120B performs as well as or better than Claude Sonnet 3.5 on the *consider* task, as measured by the accuracy metrics reported in Morin & Marttinen Larsson (2025).

Hypothesis 2: The annotation task can be performed with a significantly less extensive training process than that used by Morin & Marttinen Larsson (2025), while achieving performance comparable to or better than Claude Sonnet 3.5 according to the accuracy metrics reported in Morin & Marttinen Larsson (2025).

Hypothesis 3: DistilBERT, a small encoder-only transformer model from the BERT family, performs as well as or better than Claude Sonnet 3.5 on the same task, as measured by the accuracy metrics reported in Morin & Marttinen Larsson (2025).

4 Close replication

Morin & Marttinen Larsson (2025) present a methodological pipeline leveraging an LLM for the automatic annotation of grammatical data, based on the case study of the evaluative *consider* construction in English. Their annotation pipeline consisted of a complex “conversation” with the model over several iterative phases of training and testing. Using Anthropic’s model Claude 3.5 Sonnet (Morin & Marttinen Larsson 2025: 4), they were able to achieve an accuracy of 93% compared to the gold standard provided by a human annotator (Morin & Marttinen Larsson 2025: 6). In a first replication, we aimed to repeat Morin & Marttinen Larsson (2025)’s “replicable method” as closely as possible, using the same data and matching the training and testing to a conversation transcript the authors provided (Morin & Marttinen Larsson 2024), albeit with two open LLMs: Mistral Small 3.2 24B Instruct (MistralAI 2026) and OpenAI GPT OSS 120B (OpenAI 2026).

Table 1: Classification metrics of the three models compared to the human gold standard

Metric	Mistral	GPT	Claude (original study)
Accuracy	67%	94%	93%
Precision	77.8%	90.2%	88.1%
Recall	17.9%	94.9%	94.9%
F1-score	29.2%	92.5%	91.4%
Matthews correlation coefficient	0.25	0.88	0.86

Based on accuracy compared to the human gold standard, GPT showed comparable and even slightly better performance than Claude in the original study (GPT: 94% vs. Claude 93%), while Mistral performed considerably worse (67%). This observation is reflected in the other metrics as well: While GPT’s performance is similar or better in all evaluated metrics, Mistral fails to achieve comparable results. While Mistral’s 67% accuracy might initially sound moderately good, investigating these results as 2 x 2 matrices (see **0_Appendix.R**) reveals that it is heavily biased toward predicting ‘non-evaluative’. This shows especially in the particularly low Recall of 17.9%, since this metric answers the question how many positives of the actual positives were correctly identified.

Looking at this result, it is important to consider the distribution of *consider*-meanings in the testing dataset. The ‘non-evaluative’ classification was the majority class (63/102, 62%), and a classifier that always predicted the majority class would have achieved an accuracy of 62%. The imbalance in this dataset makes it especially important to consider other metrics when assessing model performance.

The results suggest that the (close) replicability of the approach evidently depends on the comparability of the models. Recalling RQ1, the GPT performed slightly better than Claude 3.5 Sonnet, whereas the smaller Mistral model did not. Therefore, Hypothesis 1.1 is rejected,

while hypothesis 1.2 is supported by the results. On a methodological level, we found that essential information to independently replicate the study (e.g., whether the web interface or an API was used, analysis code, etc.) was missing - forcing us to make some educated guesses as to the exact procedure.

5 Extended replication

The close replication raised questions about both the necessity of such an extensive training and testing procedure, and the need for large commercial LLMs to complete this kind of linguistic task. This led us to extend our replication to address both of these issues. First, we tested the need for the extensive ‘in-conversation’ training process outlined in Morin & Marttinen Larsson (2025). To this end, we applied a much simpler few-shot training approach with the best-performing open LLM (GPT) from our first replication using the same data.

Table 2: Classification metrics of both GPT analyses and Claude compared to the human gold standard

Metric	GPT (close replication)	GPT (reduced training)	Claude (original study)
Accuracy	94%	94%	93%
Precision	90.2%	86.7%	88.1%
Recall	94.9%	100%	94.9%
F1-score	92.5%	92.9%	91.4%
Matthews correlation coefficient	0.88	0.89	0.86

When only using the pretraining set and directly proceeding to testing afterwards, GPT achieved an accuracy of 94% (96/102). While the accuracy is equivalent to the GPT result in the close replication, the other metrics reveal some minor differences in classification performance: Recall is 100%, meaning that GPT did not miss any positives (‘evaluative’ sentences) in the extended replication. Compared to the original study, the accuracy metrics are comparable: Precision is slightly lower (GPT: 86.7% vs. Claude: 88.1%), meaning GPT predicted more false positives (‘evaluative’ for actual ‘non-evaluative’). All the other metrics are higher than Claude for the GPT replication with fewer training. Therefore, these results provide support for hypothesis 2.

Table 3: Classification metrics of distilBERT and CLaude compared to the human gold standard

Metric	DistilBERT (pretraining only)	DistilBERT (extended training)	Claude (original study)
Accuracy	62%	63%	93%
Precision	-	100%	88.1%
Recall	0%	2.6%	94.9%
F1-score	-	5%	91.4%
Matthews correlation coefficient	-	0.13	0.86

As a second extension, we replicated the annotation study using a fine-tuned distilBERT model (Sanh et al. 2020). This encoder-only transformer is considerably smaller than even a small generative LLM. The first run was similar to the reduced training approach with GPT: only the pretraining set was used for training the model, and, after using the “unsupervised training” sets for validation, we proceeded to testing. DistilBERT achieved an accuracy of 62% (63/102), matching the majority-class baseline, as it classified all test items as ‘non-evaluative’. In a second round, we combined the two unsupervised training sets and the pretraining set into one bigger training dataset, and ran the same test (see `6_BERT_extended_script.R`). In this case, distilBERT categorized one sentence as ‘evaluative’, and is still almost completely aligning with the majority baseline. For the first analysis with distilBERT, most of the metrics become meaningless when there are no positive predictions at all. Recall is 0%, because there are 39 actual ‘evaluative’ cases, but the model correctly identifies 0 of them. For the second round with the expanded training dataset, numbers could be defined because there was a positive prediction. While most of them are very low compared to Claude, the reported Precision is 100%. This metric measures how often the model is correct when it predicts positives. Since the model only predicted ‘evaluative’ in one singular case, this result is misleading. Based on both the accuracy and the further examined metrics for distilBERT, hypothesis 3 is rejected.

In summary, our results suggest that, when using a comparable open-source LLM, the training process need not be as elaborate as in Morin & Marttinen Larsson (2025) to accomplish similarly high accuracy rates. However, the smaller Mistral model and the distilBERT model showed a strong tendency toward the majority class and performed only slightly above or exactly at the majority-class baseline, indicating to not be suitable to successfully complete this kind of linguistic task.

6 Discussion

6.1 Suitability of LLMs and training process

We consider this replication successful with regard to the replicability of the methodological pipeline using an open LLM (OpenAI GPT OSS 120B), both with the iterative training outlined in Morin & Marttinen Larsson (2025) and with a reduced training process. However, both the smaller Mistral model and DistilBERT failed to achieve the desired accuracy metrics, suggesting that the approach may only be replicable with models comparable in size and functionality. DistilBERT’s performance in particular indicates that it is not well suited for this task, although this result needs to be interpreted in light of important differences between distilBERT and the other models: In addition to being a considerably smaller, encoder-only transformer, distilBERT could not be given the same type of ‘in-conversation’ instructions during pretraining as the LLMs. Thus, although the same datasets were used, the training conditions were not fully equivalent across model types.

As the results of this replication indicate, an open-source LLM can achieve performance comparable to that of a closed LLM on the linguistic classification task in Morin & Marttinen Larsson (2025). As already mentioned in the introduction, open models constitute an integral part of open research, which is why we consider this aspect very relevant when doing research with LLMs. In addition to their limited availability, closed, commercial LLMs are often developed intransparently and without ethical oversight (Liesenfeld, Lopez & Dingemans 2023). Since their training relies on vast amounts of data, including text from the internet, publicly available datasets, or proprietary sources, they are susceptible to passive privacy leakage, whereby sensitive information contained in training data may be revealed. Also, LLMs can be vulnerable to different kinds of active privacy attacks (Yan et al. 2025: 1). However, these risks affect open LLMs as well.

6.2 Replicable methods

This replication contributes to the broader discussion of replicability in linguistic research involving LLMs by examining not only whether the methodological pipeline of the original study can be implemented with different (open) models and amounts of training, but also which aspects of the pipeline and paper allow or constrain reproducibility and replicability. This is particularly important for method-focused studies using commercial LLMs, where access to the model and its underlying configurations are limited. Rather than seeing replicability as a binary outcome, our results illustrate that different components of such a paper may be replicable to different degrees.

In view of this, several aspects of the original study limit the extent to which the results can be replicated independently. Some important details, like the programming language and analysis code for implementing the pipeline, were not reported by Morin & Marttinen Larsson

(2025). In addition, the human gold standard used for evaluation seems to be established by a single annotator. It is not clear if inter-rater agreement was assessed during its creation. Such limitations leave some uncertainty about whether differences between the original and replication results are attributable to the models, implementation details, or other aspects.

Another thing that we want to address in terms of replicability is the claim that this pipeline may be applied to other tasks and other languages. While the authors already stress that the exact conditions will vary based on the task and some languages will be harder to study due to less available training data (Morin & Marttinen Larsson 2025: 7), existing research has predominantly focused on English-language contexts and paid less attention to LLMs trained on or evaluated in other languages (Yao et al. 2025: 2; Kargaran et al. 2025). Open-source models offer another advantage for linguistic research in that regard, as in some cases, pretraining data and language distribution are made transparent.

7 Conclusion

This study investigated the replicability of an LLM-based methodological pipeline for linguistic data annotation. While smaller model and distilBERT did not achieve comparable results, Morin & Marttinen Larsson (2025)’s approach was successfully replicated with one open model, and also with a significantly less extensive training approach. At the same time, it is worth noting that the study by Morin & Marttinen Larsson (2025) represents an early contribution to this specific area. Regarding the generalisability of this method, it is important to note that the grammatical construction annotated here constitutes a rather clear-cut formal case (Morin & Marttinen Larsson 2025: 7). Even with promising results from some LLMs, further exploration is needed to effectively and reliably leverage them for linguistic data annotation, especially when they are used for more complex tasks or under-researched languages.

The extensions explored in this study give insights into what can be learned from testing LLM-based methods beyond the conditions of the original study. They illustrate the importance of investigating not only which results an approach produces, but also under which conditions it remains reliable and replicable. As LLMs are increasingly used in corpus-based research, open-science practices, such as making datasets, code, model parameters, and evaluation procedures as transparent as possible, become particularly relevant.

More broadly, this study highlights the importance of replicability in linguistic research with LLMs and the advantages of using open models, which can offer at least some amount of control, and greater possibilities for reproducibility. To leverage LLMs in linguistic research, it is important to weigh their potential and their limitations, rather than assuming they can be applied to every task reliably. This also means considering broader issues associated with their usage, ranging from data rights to environmental impact.

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